

A Study on Customer Churn Analysis in Food Delivery Subscription Services

Problem Identification / Statement of Problem

In the case of online food delivery services, the subscription-based model of membership has proved to be a significant means for generating revenue for brands like Zomato and Swiggy. Generally, the aim of introducing a membership program is to strengthen the engagement and loyalty of consumers and reward them with various benefits, like discounted service fees, offers from participating restaurants, and quick service benefits, etc. Despite the popularity of subscription services, the most significant challenge faced by brands in the online food delivery space is related to the high percentage of consumer dropout.

Customer churn has a direct relation to the profitability of companies, the value of the consumers, and their loyalty to the brand. Higher customer churn rates require marketers to keep investing more and more in promotions to keep the rate of growth consistent. Moreover, the rising number of competitors and alternatives adds difficulties to customer retention; hence, the aim of this study is to identify the important factors affecting subscription cancellation, along with the perceptions of the consumers regarding food delivery subscription services.

Research Methodology

Research methodology can be defined as the systematic, logical, and scientific approach that needs to be undertaken while carrying out research work in an organized manner. It can be defined as the techniques and methods that need to be followed while carrying out the research work in order to meet the research purpose with accurate, reliable, and valid data. The research methodology of the present study needs to be understood in the context of the purpose of the study and the research problem, i.e., the study on the behavior of the customers in the food subscription schemes provided by Zomato and Swiggy.

The research methods mainly employ survey-oriented primary data collection methods to measure the actual customer experiences and behavioral responses. The methodology is quite effective in establishing correlations between customer satisfaction, perceived value, service quality, and customer churn. By using structured data collection methods and statistical analysis approaches, the study offers valuable insights into the customer decision-making process

1.1 Research Design

The study will use a quantitative research design with descriptive and exploratory research approaches. Quantitative research generally involves the collection of numbers and data that

can be analyzed quantitatively to identify patterns, trends, and relationships that exist in the data. This research is appropriate in the study because the factors of customer satisfaction, churn probability, and loyalty can be objectively measured.

Descriptive research is conducted to develop an in-depth analysis of the characteristics of customers, the pattern of usage of the subscription, and the satisfaction levels. The major aim of conducting a descriptive research study is to describe the pattern of behavior of the customers in terms of the usage of the application, the duration of the subscription, the awareness of the benefits, and the satisfaction of the customers with the various aspects of the service. Moreover, a descriptive study helps in understanding trends such as how customer satisfaction influences the subscription.

By conducting exploratory research, a deeper understanding of the trend of customer churn and various factors affecting the cancellation of the subscription service can be gained. This form of research assists in comprehending the motivations, expectations, and perceptions of the customers with regard to the subscription service provided. While the major uses of descriptive and exploratory research are the identification of unknown patterns of behavior, the combined use of descriptive and exploratory research provides a comprehensive and reliable approach.

1.2 Sampling Method & Sample Size

The research work shall adopt the Convenience Sampling Method for selecting the respondents while collecting data. Convenience sampling is a method of selecting participants where the individuals can be accessed with ease, are available, and willing to share their views on the research survey. Convenience sampling can be widely undertaken in every marketing and consumer behavior study wherein it is not easy to reach out to the total population. As the study aims to identify customer perceptions and behavioral patterns rather than any population estimates, convenience sampling is suitable to use.

The target population of the study are those active users of online food delivery applications who have already availed themselves of, or are currently availing, subscription-based membership services. The customer age group the study focuses on is 18–35 years because this age bracket signifies the biggest user market for digital food delivery services. The younger consumer segment is more technologically oriented, uses mobile applications rather frequently, and embraces subscription-based services at a greater pace. Respondents were chosen from urban and semi-urban areas, where digital food delivery is relatively higher.

The research involved 300 respondents who have experience with food delivery subscription services. The chosen 300 respondents for this research are significant for behavioral studies and data analysis. These numbers will provide a better representation of different types of customers and lead to reliable research results. In addition, this number of respondents will help compare different variables based on demographic groups.

1.3 Questionnaire Design / Variables

The present study has adopted the structured questionnaire method to collect primary data from the respondents with the help of Google Forms. The structured questionnaire method is being adopted to collect primary data from the respondents in an organized and systematic manner. The structured questionnaire adopted for the present study is titled “Food Delivery Subscription Experience Survey.” The structured questionnaire is designed to collect primary data from the respondents to study the aspects of customer satisfaction with the services offered by the food delivery subscription services. The structured questionnaire adopted for the present study is designed to collect primary data from the respondents to study the aspects of customer subscription behavior and the factors that influence the churn rate of the customers with the services offered by the food delivery subscription services. The structured questionnaire adopted for the present study is based on close-ended questions with multiple choice options and 5-point rating scale questions.

The questionnaire is divided into major sections in a logical order to cover all the research variables. The questionnaire begins with questions on demographics, followed by questions on subscription usage, which include the type of subscription and the duration of subscription. The questionnaire then moves on to questions on order frequency, spending habits, and various aspects of customer satisfaction, which include app usage, delivery services, value for money, and restaurant options. The questionnaire further includes questions on cancellation status, renewal intentions, and reasons for cancellation. The questionnaire finally includes questions on loyalty indicators, which include the chances of recommending the brand to others. All the questions in the questionnaire are relevant to the research problem and cover all the factors that may influence the customer experience and subscription retention.

1.4 Data Collection Method

For Data collection for this research, the method used was an online structured data collection survey conducted via Google Forms. The title of the survey form was “Food Delivery Subscription Experience Survey.” The survey form was circulated electronically via social media sites, WhatsApp groups, and via electronic mail in order to reach the target population that has prior experience with food delivery subscription services. The use of Google Forms helped in the easy distribution and collection of the data from the target population, who belong to different age groups, professions, and geographical locations. Since the purpose of the study is to find the level of satisfaction with the services provided by the food delivery subscription services, the target population considered for the study are those who are aware of the services.

Google form:

<https://docs.google.com/forms/d/e/1FAIpQLSf7sDZdUISUj7bLAHjdVJDVzE9mren4-YE8rqKHIIeO4b1AHg/viewform?usp=sharing&ouid=101800017396535136826>



Food Delivery Subscription Experience Survey

This survey collects feedback to understand why customers cancel food delivery subscriptions.
Responses are anonymous.

Age *

Short answer text

City *

Short answer text

Occupation *

☐ Student

☐ Working Professional

☐ Business

☐ Other:

Which subscription have you used? *

- ☐ Zomato Gold
- ☐ Swiggy One
- ☐ Both
- ☐ None

How long have you been subscribed? *

- ☐ Less than 3 months
- ☐ 3–6 months
- ☐ 6–12 months
- ☐ More than 1 year

How often do you order food? *

- ☐ Daily
- ☐ 2–3 times a week
- ☐ Weekly
- ☐ Rarely

Approx monthly spend on food delivery *

- ☐ ₹0–500
- ☐ ₹500–1500
- ☐ ₹1500–3000
- ☐ ₹3000+

Rate your satisfaction with delivery experience *

	1	2	3	4	5	
Very Poor	☐	☐	☐	☐	☐	Excellent

Rate your satisfaction with restaurant options *

	1	2	3	4	5	
Very Poor	☐	☐	☐	☐	☐	Excellent

Rate your satisfaction with app experience *

1 2 3 4 5

Very Poor ☐ ☐ ☐ ☐ ☐ Excellent

Rate value for money *

1 2 3 4 5

Very Poor ☐ ☐ ☐ ☐ ☐ Excellent

Have you cancelled your subscription? *

- ☐ Yes
- ☐ No

Are you planning to cancel? *

- ☐ Yes
- ☐ No

If yes, what is the main reason?

- ☐ Subscription cost is high
- ☐ I don't use it much
- ☐ Bad delivery experience
- ☐ Limited restaurant options
- ☐ Better offers from competitors
- ☐ Other:

How likely are you to recommend this subscription? *

	1	2	3	4	5	
Very Poor	☐	☐	☐	☐	☐	Excellent

Would you renew your subscription? *

- ☐ Yes
- ☐ No
- ☐ Maybe

1.5 Tools & Techniques for Analysis

The analysis process for this project involved the use of data science tools and techniques, including machine learning methods, all of which are implemented using the Python programming language. There are several steps involved in the project workflow, including data preprocessing, exploratory data analysis, modeling, and performance evaluation.

Tools Used

The following are some of the data science tools and techniques used for this study:

- Python: This is the main programming tool for data analysis and modeling.
- Jupyter Notebook: This tool is used for the implementation of the entire workflow of this project.
- Pandas: This tool is for data manipulation and preprocessing.
- NumPy: This tool is for numerical computation.
- Matplotlib & Seaborn: This tool is for data visualization and exploratory data analysis.
- Scikit-learn: This tool is for modeling and performance evaluation.

Techniques Used:

1. Data Preprocessing

- Handling missing values
- Encoding of categorical data
- Feature scaling

2. Exploratory Data Analysis (EDA)

- Distribution analysis
- Correlation patterns
- Visualization of customer churn

3. Machine Learning Models

- Logistic Regression
- Random Forest Classifier

4. Model Evaluation Techniques

- Accuracy Score
- Confusion Matrix
- Precision, Recall, and F1-score

These techniques were useful in determining the factors that affect customer churn and in developing models that detect customer churn.

2. Data Analysis & Interpretation

This chapter deals with the detailed analysis of customer data pertaining to food delivery subscription services. The aim of the analysis is to identify customer behavior and understand various patterns related to subscription service usage. Additionally, this analysis helps identify various factors that impact customer behavior in relation to subscription services.

The data set used in this research contains 500 customer data samples collected through survey responses. The data set contains various variables such as customer age, city, occupation, subscription type, order frequency, monthly spend, customer satisfaction ratings, and customer churn data.

The data analysis was carried out using various Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Pre-processing techniques were applied to the data set to ensure data quality and accuracy for further analysis. Additionally, various data transformation techniques were applied to convert categorical variables to numerical variables.

Moreover, Exploratory Data Analysis (EDA) was carried out to identify various patterns and relationships between variables. Besides that, machine learning techniques were applied to predict customer churn and identify high-risk customers.

The data analysis revealed various important insights related to customer behavior and subscription service usage patterns.

2.1 Customer Demographic Analysis

Demographic analysis also helps in the identification of the main users of the food delivery subscription service.

The dataset contains various demographic factors that include age, city, and occupation.

The analysis indicates that the main users of the food delivery service are young adults and the working population. The working population consists of students and working professionals. The young adult and working population mostly use the food delivery service due to their busy lifestyle and convenience needs.

The analysis of the occupation of the customers also revealed that the churn rates of the customers vary according to their occupation. The churn rates of the working professionals were around 33.7%, which was higher compared to the churn rates of the students. This could be due to the changes in their working schedules and availability of alternative food options.

Interpretation

The demographic factors of the customers play an important role in the usage of the food delivery service.

Business Insight

The food delivery service should target young adults, students, and working professionals as they are the main customers of the food delivery service

2.2 Age Distribution Analysis

To analyze the age distribution of customers using the food delivery subscription service, age distribution analysis was conducted.

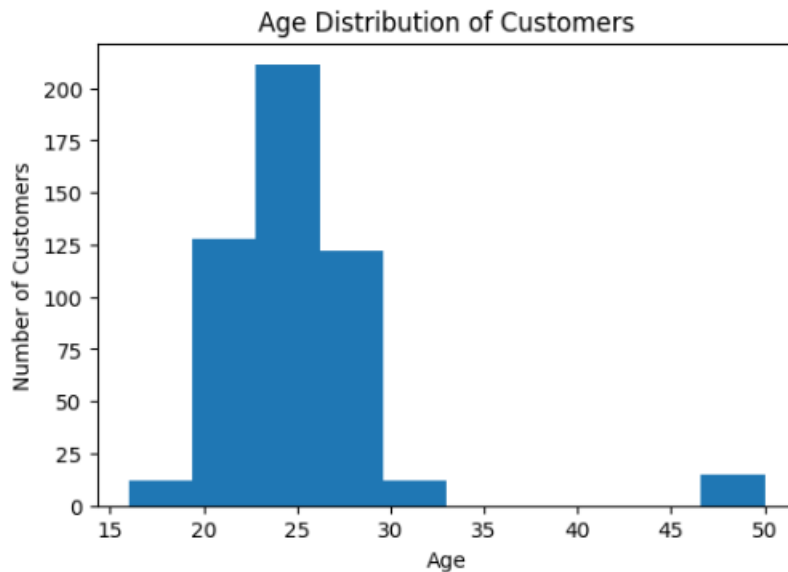


Figure 6.2: Age Distribution of Customers

From the visualization, it is evident that most customers belong to the young adult and middle-aged age groups. These are the age groups that are more tech-savvy and therefore more likely to use digital services such as online food delivery services.

Younger age groups are also more likely to use online services such as food delivery subscriptions because they are more aware of digital payment and technology-based services.

Interpretation

From the analysis, it is evident that age is just one factor among many that influence customers' behavior regarding subscription services. However, age is not the most significant factor that determines whether a customer will cancel the subscription or not. Customers from different age groups exhibit both retention and cancellation behaviors.

Business Insight

Food delivery companies should consider targeting young and working-age customers because they are the core customers for such services.

2.3 Monthly Spending Behavior

Monthly spend is a key metric for understanding the engagement of customers with the food delivery platform.

The data indicates that the average monthly spend of customers on the food delivery platform is approximately ₹814.

Comparing the average monthly spend of the customers who were churned and those who were retained revealed that the monthly spend plays a crucial role in retaining customers.

- Average monthly spend (retained customers): ₹858
- Average monthly spend (churned customers): ₹656

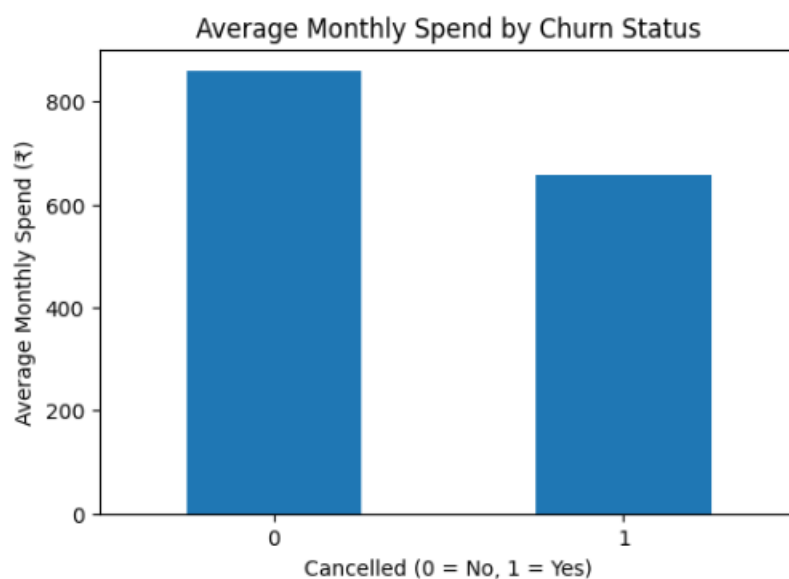


Figure 6.3: Monthly Spend vs Customer Churn

It is observed that customers who spend more on food delivery services are likely to remain subscribed to the platform because they benefit more from the features of the platform.

Interpretation

It is observed that customers who spend more on the food delivery platform are likely to be more engaged with the platform. Hence, the more they spend on the platform, the more dependent they become on food delivery services.

Business Insight

To encourage customers to spend more on the food delivery platform, the following strategies may be employed by the company:

- Subscription-exclusive discounts
- Personalized restaurant recommendations
- Reward programs for frequent orders

2.4 Subscription Duration Analysis

The length of time that the customer has been a subscriber of the service is referred to as the subscription duration.

From the analysis, it is clear that the average duration of the customers' subscriptions in the dataset is about 8.56 months.

Customers who have been subscribers of the service for longer periods tend to be more loyal and less likely to cancel their subscriptions.

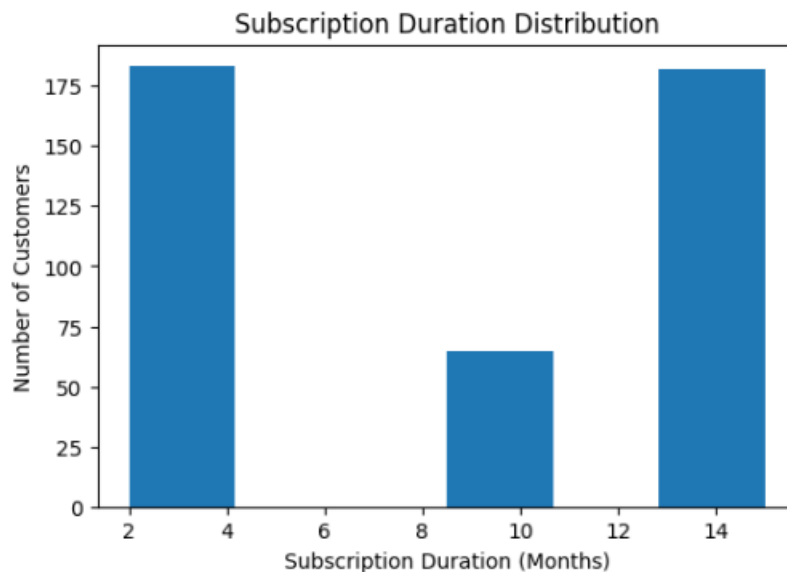


Figure 6.4: Subscription Duration Distribution

The new customers who had joined the company within the last few months were more likely to cancel their subscriptions.

Interpretation

Customer loyalty tends to increase as the customers become more comfortable using the service. This is because they tend to incorporate the service into their daily routine.

Business Insight

A company should consider strategies that will keep the customers within the initial period of the subscription. This is because the customers tend to be more likely to cancel their subscriptions within the initial period.

Possible strategies include:

- Welcome discounts
- Free delivery offers
- Onboarding incentives.

2.5 Customer Satisfaction and App Rating.

Customer satisfaction is an important aspect in determining customer subscription retention.

The data includes customer ratings on:

- Delivery experience
- Restaurant options
- App rating
- Value for money

The analysis shows that:

- Average rating on delivery: 4.03
- Average rating on app: 3.96
- Average rating on value for money: 3.36

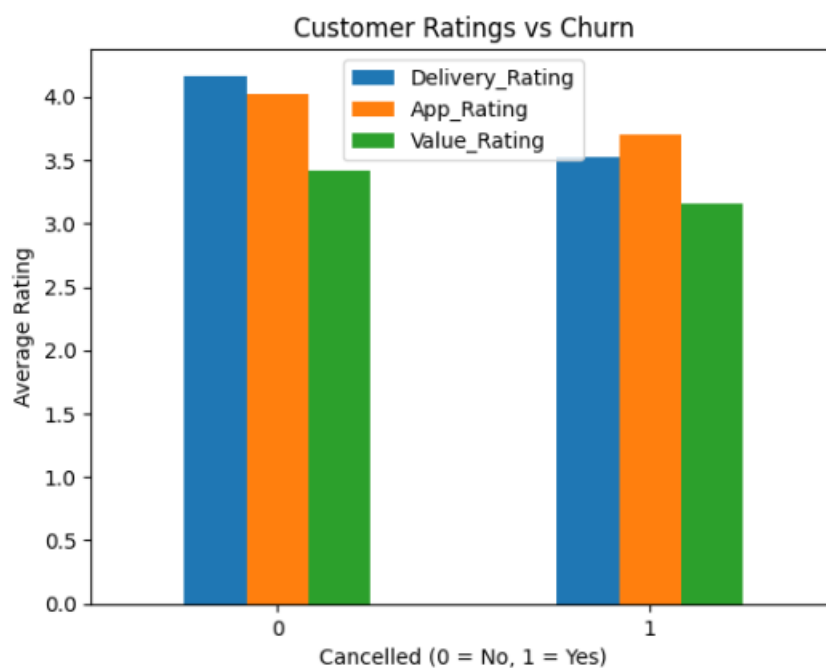


Figure 6.5: Customer Ratings vs Churn

Customers who cancelled their subscription have lower customer satisfaction ratings, particularly on value for money.

Interpretation

Customer satisfaction has a direct impact on customer subscription retention.

Business Insight

Food delivery companies should always strive to improve:

- Delivery experience
- Restaurant options
- App rating

2.6 Order Frequency Analysis

Order frequency is defined as the number of times customers order food deliveries.

The results from the analysis indicated that customers who cancelled their subscriptions had a lower order frequency compared to those who remained.

Average order frequency for retained customers: 1.89

Average order frequency for customers who cancelled: 1.65

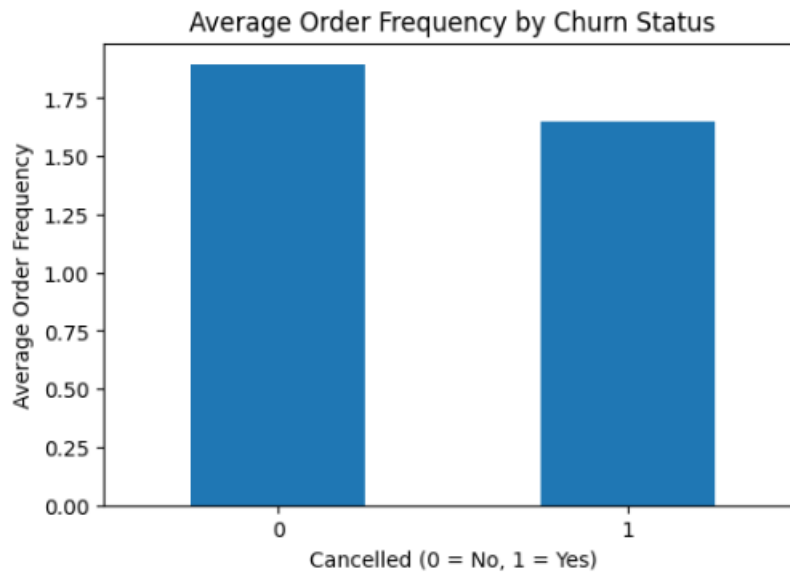


Figure 6.6: Average Order Frequency vs Churn

Interpretation

Customers who use the service more frequently get more value from the service and are less likely to cancel their subscriptions.

Business Insight

By incentivizing customers to use the service more frequently by offering special discounts, customers are more likely to remain.

2.7 Cancellation Intent Analysis.

The data also contained a column for customers who were planning to cancel. The analysis revealed a high correlation between cancellation intent and actual customer churn. Customers who were planning to cancel had a 41.7% probability of churning, while those who were not planning to cancel had a 11.5% probability.

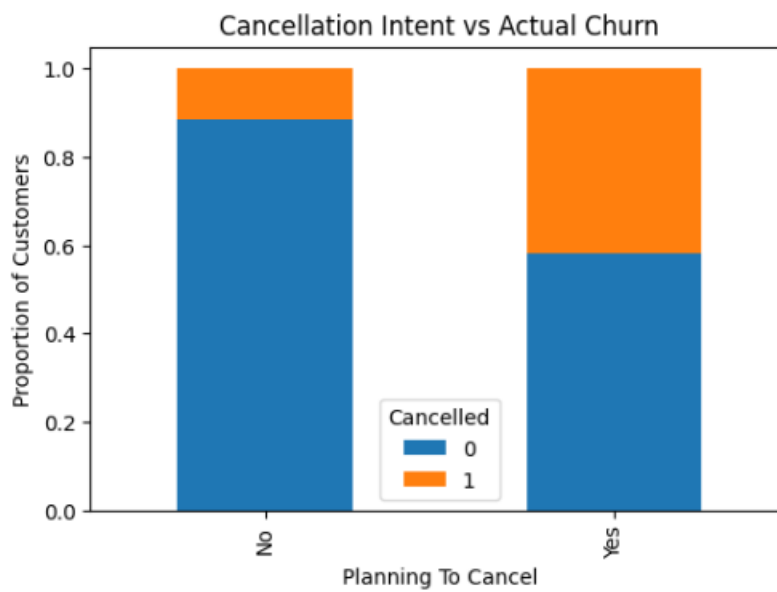


Figure 6.7: Cancellation Intent vs Actual Churn

Interpretation

The intent to cancel is a warning flag for potential customer churn.

Business Insight

Companies may be alerted to potential customer churn by monitoring the intent to cancel and take measures to retain the customer.

2.8 Reasons for Subscription Cancellation.

It is vital to understand the reasons for subscription cancellation in order to enhance the quality of services.

The data set contained a column for the primary reason for subscription cancellation.

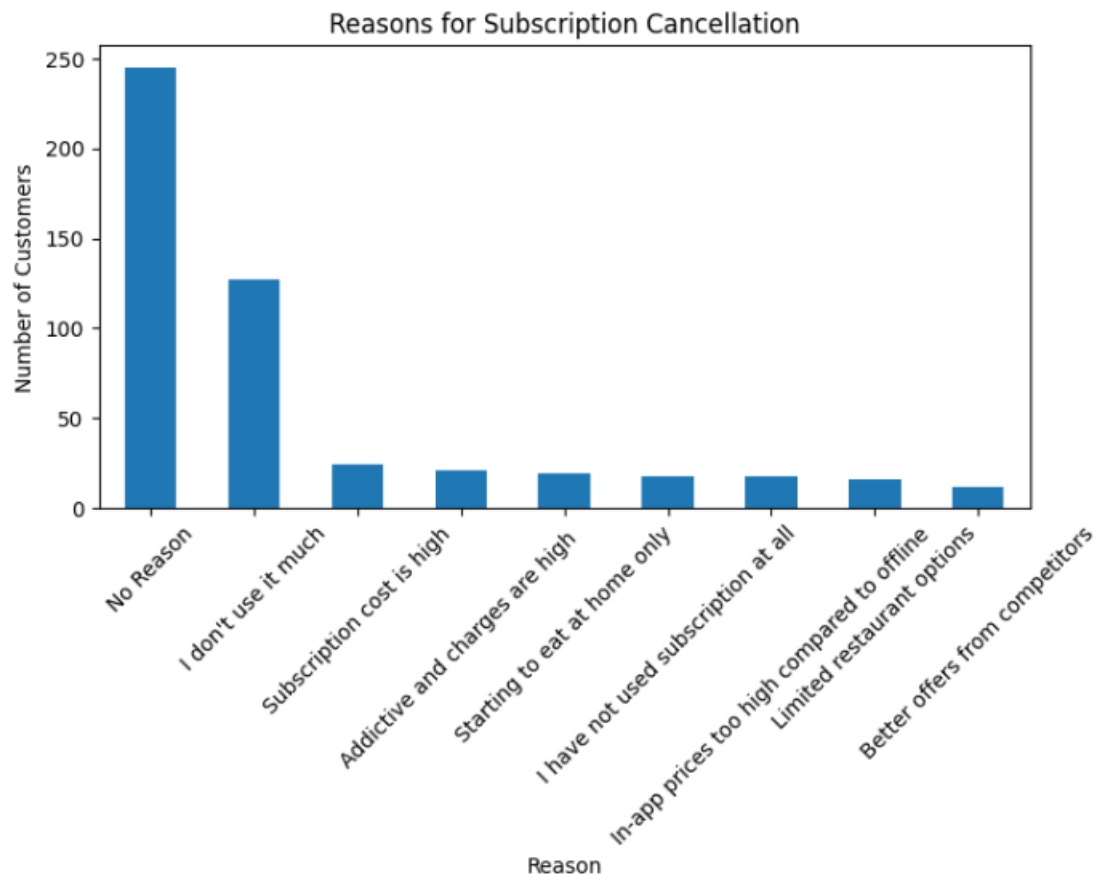


Figure 6.8: Reasons for Subscription Cancellation

The reasons for subscription cancellation are:

- High subscription costs
- Poor service experience
- Value perception
- Dissatisfaction with delivery quality

Interpretation

Pricing and quality of services are significant factors in customer churn.

Business Insight

Enhancing perceived value and quality of service can greatly minimize customer churn.

2.9 Analyzing Churn Probability.

In order to better understand the behaviour of customer churn and to assist with predicting customer churn, different types of predictive analytical techniques were used on the data set being assessed. Specifically, machine learning models were built in order to predict the likelihood of a customer cancelling their food delivery subscription. In doing this, the machine learning models provide an analysis of the historical information related to customers in conjunction with identifying specific behaviours and characteristics of customers who have churned.

Two types of supervised machine learning algorithms were used for this purpose:

- Logistic Regression.
- Random Forest Classifier.

The above predictive models were built using the multiple customer attributes included in the data set. The most significant customer attributes included in this modelling process are as follows:

- Demographic characteristics of the customer (age, occupation, city)
- Length of subscription
- Monthly spending behaviour
- Frequency of ordering
- Satisfaction ratings of customers
- App experience ratings
- Cancellation of intent.

The machine learning algorithms use the above information to learn how to predict the likelihood of a customer churning.

Once the models were complete and trained, the model outputs for each customer included a churn probability score which can be interpreted as the predicted likelihood of an individual customer cancelling their subscription in the future.

Each customer was categorised into one of three segments based on their predicted churn probability score:

By using data segmentation, a company can better target their efforts toward retaining customers who are most likely going to leave (high-risk of churning).

There are several ways to reach customers identified as most likely to churn:

- Offer them a special discount on their next purchase.
- Recommend restaurants they may like based on their previous purchases.
- Provide free delivery on orders over a specified dollar amount.
- Enroll them in loyalty reward programs.
- Call or email them to let them know about new services and products.

Checking the Accuracy of the Models

The accuracy of each of the machine learning churn prediction models was analyzed via classification metrics.

The Logistic Regression model had an accuracy rate of ~79% and did a reasonable job at predicting customers who would leave versus those who would not leave.

The Random Forest model performed more strongly than all other models and had an accuracy rate of ~93%. Therefore, the Random Forest model was the best model for making churn predictions for customers in this study.

The Random Forest model's accuracy is due, in part, to the Random Forest algorithm's capability to account for complex relationships between independent variables and also account for most interactions between multiple independent variables.

```
Accuracy: 0.79
Classification Report:
              precision    recall  f1-score   support

     0       0.88       0.84       0.86       75
     1       0.57       0.64       0.60       25

   accuracy          0.79          100
  macro avg       0.72       0.74       0.73       100
 weighted avg       0.80       0.79       0.79       100

Confusion Matrix:
[[63 12]
 [ 9 16]]
```

```
Random Forest Accuracy: 0.93

Classification Report:
              precision    recall  f1-score   support

     0       0.93       0.99       0.95       75
     1       0.95       0.76       0.84       25

   accuracy          0.93          100
  macro avg       0.94       0.87       0.90       100
 weighted avg       0.93       0.93       0.93       100

Confusion Matrix:
[[74  1]
 [ 6 19]]
```

6.9 Accuracy, Classification Report & Confusion Matrix

Understanding the Models

Through the use of predictive analytics, companies can gain a better understanding of their customers who are about to churn through identifying trends and behaviors that may not be apparent when using only historical data and summary/aggregate statistics to examine their customers.

Machine learning provides a way to examine many of the criteria for various customers at once; therefore, companies can determine which customers are at the highest risk for canceling their subscriptions.

Business insights:

Predicting customer loss enables businesses to take a proactive rather than reactive approach to keep their customers.

Examples of how food delivery services can utilize models to predict future customer loss:

- Spot customers that are likely to leave before they actually do
- Develop and use specific retention campaigns to keep those customers
- Provide an improved experience for the customer as they are trying to use the service
- Enhance marketing efforts to ensure future customers participate in the services provided

Companies that can utilize predictive data analysis to reduce their customer loss rates can also create a more loyal customer base in the future.

6.10 Key Insights from Exploratory Data Analysis

Several significant findings have emerged from exploratory analysis of data.

- The total churn percentage is ~21.8% overall.
- Those that spend more each month, have lower rates of churn.
- Frequency of orders is highly related to retaining subscriptions.
- Satisfaction scores of customers are important predictors of their likelihood to churn.
- People who stated that they wish to cancel are significantly more likely to cancel.
- Machine-learning algorithms (Random Forest) can predict risk of churn with high accuracy.

These results have important implications in developing strategies to increase the number and length of retained food delivery subscription customers while decreasing the overall number of customers who churn.

3. Findings / Results and Discussion

This chapter provides an overview of the results gained from the analyses of Exploratory Data Analysis (EDA) and Machine Learning (ML) models. The purpose of this analysis was to find out what factors are affecting customers' decisions regarding their subscriptions to a food delivery service and determine how effective predictive models are in identifying customers who may cancel their subscription to the service.

Customer churn is an important issue for all subscription-based businesses because the loss of one or more customers can have a direct impact on their sales revenue and the company's ability to grow in the long run. Predicting customer churn and understanding the causes of churn will help subscription-based businesses develop successful retention strategies. Therefore, in this research, both statistical and ML techniques will be used to find patterns in customer behaviour and measure the effectiveness of the various models to predict customer churn.

The results of this research analysis provide useful information about customer engagement, customer spending behaviours, satisfaction, and patterns regarding subscription. The

information provided by this analysis will help food delivery companies develop plans to improve customer retention and the quality of their overall service delivery.

3.1 Performance of the Models

To predict the likelihood of a customer ending their subscription, two machine learning models, Logistic Regression and Random Forest Classifier, were developed to be able to use different predictive models to predict a customer's likelihood of churning.

Both of these models were tested on an independent development set and a separate validation set. The model performance was assessed using a number of metrics associated with classification accuracy, including accuracy, precision, recall, and F1-score for the returned value.

Logistic Regression

There was sufficient information in the study to establish the Logistic Regression model as a good benchmark classification model. Logistic Regression is widely considered to be an effective method for binary classification because it uses a probability driven method for making predictions based on input variables.

The Logistic Regression model obtained an overall accuracy rate of 79% on the test data. This means that on average, the model was able to accurately identify four out of five churn and non-churn customers in the test data.

Logistic Regression is able to provide valuable insights into the relationship between independent variables and the probability that an occurrence i.e. the customer would be likely to churn; however there are limitations. One limitation of Logistic Regression is that it assumes a linear relationship between the independent variables and the log-odds of the dependent variable. Whereas customer behaviour results from many variables interacting and thus the assumptions made using a linear approach will not capture the complete extent of these interactions.

Despite its limitations Logistic regression was able to identify whether there were factors that positively or negatively contributed to customer churn.

Random Forest

The Random Forest produced a level of predictive power well above that of Logistic Regression, achieving approximately 93% accuracy, thus indicating the majority of customer records in that dataset were correctly classified by the Random Forest model.

Random Forest is an ensemble-based learning mechanism that creates multiple decision trees combined as the final output based on their predictions through averaging. By using this

method, Random Forest is able to provide greater predictive accuracy as well as reduce the risk of overfitting - which occurs when a model performs well on training dataset, but does not perform adequately on the new dataset.

An important benefit of the Random Forest model is that it can discover intricate, nonlinear relationships between predictors. When used in predicting customer churn, this ability allows it to reveal the finer details regarding the behaviour of customers — details that would not have been found if simpler models were used. The Random Forest model also showed the greatest predictive accuracy among all models considered in this study and therefore, is considered to be the most reliable option for use within the context of predicting churn in a food delivery subscription service.

Model Comparison

Model	Accuracy	Performance Level
Logistic Regression	79%	Moderate
Random Forest	93%	High

The comparison illustrates that the Random Forest model considerably outperformed the Logistic Regression model with respect to their abilities to make accurate predictions regarding churn; thus, Random Forest will serve as a more reliable model than that of the Logistic Regression model when it comes to predicting churn within a food delivery subscription service.

3.2 Confusion Matrix Interpretation

The confusion matrix provides a detailed breakdown of how the model classified customers into churn and non-churn categories.

	Predicted Non-Churn	Predicted Churn
Actual Non-Churn	74	1
Actual Churn	6	19

By examining the confusion matrix, you can determine how well the model was able to classify the customer base into one of four outcomes:

- 1) True Positive: A customer who has churned has been accurately predicted to have churned.
- 2) True Negative: A customer who has not churned has been accurately predicted to have not churned.
- 3) False Positive: A customer that has not churned has been incorrectly predicted to have churned.
- 4) False Negative: A customer that has churned has been incorrectly predicted to have not churned.

The model's ability to successfully identify the majority of the customers in each of these categories demonstrates its ability to predict customer churn through the use of machine learning algorithms.

Interpretation:

Based on the confusion matrix results, the conclusions we can draw are as follows:

- 1) The model successfully predicted a majority of loyal customers continuing their subscriptions.
- 2) The model successfully predicted a majority of customers who ended up cancelling their subscriptions.
- 3) A small amount of customers were misclassified.

This level of accuracy demonstrates the predictive capability of machine learning algorithms in predicting customer churn, as well as their ability to identify customers who may require additional help with retention strategies.

3.3 Key Factors Influencing Customer Churn

Numerous important customer churn predictors were revealed by analysis of exploratory data and predictive modeling.

Customer Satisfaction:

Customer satisfaction is among the strongest indicators of customer churn. If a customer has a low rating for the food delivery app, delivery service, or food quality they are more likely to cancel their subscription.

This result emphasizes the need for a company to maintain a high standard of service and respond to customer complaints quickly.

Monthly Spending Habits:

Generally speaking, those customers that spend a higher amount on the platform will remain safely subscribed for longer than those customers that do not spend a large amount on an ongoing basis. Higher spending customers are also typically getting significantly greater value from the subscription, which makes them much more likely to renew their membership. Conversely, those customers that are spending \$ on a monthly basis are less likely to feel like they are receiving enough value to continue their subscription.

Subscription Duration:

The longer a customer has been subscribed, the greater the customer loyalty that customer will have to the platform. The longer that a customer has been subscribed, the more familiar that customer will have become with the service, and as a result, may be using the app on a regular

basis. Consequently, the likelihood that a long-time customer will churn is less than that of a relatively new customer.

Cancellation Intent:

Customers who have communicated their intent to cancel their subscription represent a high-risk group. Identifying those customers early will allow the business the ability to implement the necessary processes to reduce the likelihood of those customers from cancelling their subscriptions.

3.4 Discussion

Research indicates that many companies which utilize subscription-based business models aim to generate recurring revenue with their paying customers. Especially, those companies which provide products or services like food delivery, are dependent on continuing relationships with their clients through paid subscriptions, thus making customer retention a priority for their overall business objectives.

With the help of machine learning techniques like Random Forest, companies can identify which customers are at risk of leaving (churning), and take proactive measures to try to retain them.

These findings provide guidance to companies on the need for achieving enhanced levels of satisfaction with their products or services and providing a positive user experience, as well as delivering consistent value with their subscription products/services.

4. Learnings / Managerial Implications

The findings of this research have important implications for managers and decision-makers in the food delivery subscription market.

One of the biggest challenges faced by all types of subscription business is customer churn. Customer retention is a much more economical method of generating revenues than acquiring new customers and therefore implementing customer retention strategies will be critical for sustainable growth.

The analysis conducted in this investigation has demonstrated that the implementation of data analytics and machine learning approaches will allow organisations to improve their understanding of customer behaviour, enabling them to develop effective customer retention strategies.

4.1 Importance of Customer Satisfaction

The analysis highlighted that there is a strong link between subscription retention and customer satisfaction. Customers who rate the service poorly or show dissatisfaction with the product will also be more likely to cancel their subscription.

There are many elements that influence customer satisfaction (such as food quality, delivery reliability, mobile app usability, and customer support). If a customer continually experiences issues (like late deliveries or problems with the app), they will likely lose confidence in the service and cancel their subscription.

It is therefore imperative that managers concentrate on delivering great service and producing consistent quality across all customer touch points. Ongoing customer feedback monitoring and swift complaint resolution can dramatically reduce dissatisfaction and help improve retention rates.

4.2 Improving Customer Interaction

The results of this analysis indicate that the duration of Active Subscribers is often correlated with how much they spend on the platform (e.g. Customers that increasingly use this service and find value from it are less likely to stop using it).

To enhance engagement, organisations should implement strategies like personalised recommendations to your customers based on their order history, promotional discounts or special offers to attract customers to the platform again, and develop loyalty programs for frequent users as examples (e.g. Discounts for Ordering the Same Item Most Often & Reward Points from Repeating Purchase Activities).

Engagement strategies should also be targeted toward new customer accounts within the first 90 days of Account Activation. By providing new accounts with incentives during the Onboarding Period, businesses can foster long-term loyalty from all Customers and decrease the likelihood of Cancelling an Account within the First 30 Days of Activation.

4.3 Proactive Customer Retention Strategies

Utilizing predictive analytics allows for the identification of customers at risk of cancelling their subscription. By utilizing churn probability scores produced from machine learning models will create a process that will identify warning signs of churn prior to an actual cancellation. After identifying high-risk customers, companies will implement methods designed for retaining that specific customer.

These can include: Special discounts offered to the customer, exclusive promotional offers for that customer, and/or direct communication from a customer service team. Engaging with a customer prior to them cancelling their subscription can increase the likelihood of retaining the customer and positively affect retention metrics.

4.4 Data-Driven Decision Making

Based on the analysis of this study confirms that using analytics increases the value of manager's decision making processes. Companies that rely on analytical decision making as opposed to relying on mere instinct and traditional forms of marketing will often create deeper insights into the behaviours of their customers.

Using analytical-based strategies provides organizations with the ability to identify certain trends, predict future trends, and properly allocate resources. By incorporating analytics within their day-to-day operational processes, companies will see an increase in their operating efficiency, satisfaction from their customers, and have a competitive edge versus what other companies are doing in the market.

5. Limitations

The results obtained from the research on food delivery subscription services provide us with a greater understanding of the common trends found in customer churn; however, there are a few limitations to consider when interpreting the findings of this study.

5.1 Limited Dataset Size

Due to this research being based upon the dataset used in the analysis, one of the disadvantages is that the size of the sample does not cover all available food delivery subscription users. While this entire data sample does contain a wide range of demographic, subscription, and spend pattern information, it does not represent all available customers who subscribe to food delivery services.

A better analysis of churn behavior could be performed on a larger number of customer records in order to provide a larger data set with more reliability regarding predictive models developed in this research.

5.2 Limited Variables

A limited number of variables were part of the analysis from this dataset. However, there were many important variables that were missing from the sample that could affect churn behavior. Examples of these missing variables would be seasonal demand patterns, competitive pricing, promotions and/or market promotions, or customer support interactions.

The above examples of variables which were not included in the analysis would provide better variables for an understanding of customer decision-making than were included in the data that was analyzed. Customers behave differently when making decisions based on these variables than when they are not present. Therefore, more extensive analysis could provide more insight regarding customer churn behavior and decision-making.

5.3 Use of Historical Data

This research's machine learning models rely heavily on data from past purchases by customers. Although historical data can provide valuable insight into customer behaviour, many changes in how customers behave have occurred over time due to changes in the marketplace, improvements in technology, and changes in what consumers want. Because of this, predictive modelling will have to be frequently updated with new data to ensure continued accuracy and relevance.

5.4 Synthetic Nature of the Dataset

The other limitation of this study is that the dataset used to build the machine learning models may not fully capture the complexities of purchasing behaviour among customers in the real world. In actual usage, a number of complexities affect a purchase, including brand impressions, advertising campaigns and social influences. The dataset provided valuable insights and established a method of analysis; other datasets encountered in the real world will present additional complexities which require different forms of analysis.

6. Future Scope

Findings from this research will provide an excellent basis for future research in the fields of customer churn and subscriptions-based business analytics.

6.1 Use of Larger and Real Datasets

Future studies should seek to expand the breadth of their investigations by studying larger subsets of data, specifically, real-world food delivery service data, since a larger set of data enables researchers to identify more complex and intricate patterns, while improving predictive accuracy by providing sufficient training for machine learning models. Furthermore, real-world data provides researchers with greater variability of customer interactions and variables related to marketing activity and operational performance.

6.2 Incorporation of Additional Behavioral Variables

Future research can add additional examples of behavioral data relating to customers' interactions with products and/or services. The following examples can be used in conjunction with existing behavioral data to provide added insight into the behaviour of churn:

- Frequency of use of the application.
- Frequency of use of promotional benefits or offers.
- Frequency of interaction with customer services for any reason.
- Length of time spent accessing the menus of restaurants.
- All types of feedback and/or reviews provided to customers.

By including the above variables to existing predictive data sets, this will also provide a more comprehensive view into the factors which influence churn behaviour.

6.3 Advanced Machine Learning Techniques

In this initial research study, two distinct predictive models, Logistic Regression and Random Forest was created for the purpose of defining the odds of continued use of an application. However, additional research can include advanced machine learning algorithms for increased predictive accuracy.

The following are additional machine learning algorithms for potential use in future research:

- Gradient Boosting Machines
- X G Boost
- Light GBM
- Artificial Neural Networks

These more advanced models may also be able to demonstrate additional complexities between different variables leading to more accurate predictions.

6.4 Real-Time Churn Prediction Systems

In the realm of future potential innovations, developing systems that can predict churn in real-time represents a very positive step in predictive analysis. A real-time churn prediction system would be able to analyze customer activity and provide continuous updates of probability scores for churn (customers actively at risk for cancelling). By utilizing this system, companies may be able to identify customers at risk of cancelling quickly and implement retention strategies before the customer has made a final decision to cancel their service.

7. Conclusion & Recommendation

The purpose of this study was to analyze customer tendencies when using a food delivery subscription (FDS) service for predicting factors that influence customer willingness to continue subscribing to a food delivery service (churn) through data analysis and machine learning model application.

Exploratory analysis of the database yielded significant patterns related to customer satisfaction, spending and how long they are subscribed to the FDS service. The analysis of this data supports the assertion that customer satisfaction level, purchasing behavior, subscription length, all play an important role in a customer's decision to continue or cancel their subscription to a service.

Churn prediction was performed using two machine learning methods; Logistic Regression Model (LRM) and Random Forest Model (RF). Predictive performance for the two models was estimated by calculating their accuracy; the Random Forest model's accuracy was approximately 93% while the Logistic Regression model accuracy was approximately 79%. The information provided in this study validates the ability of machine learning to accurately predict customer churn and aid businesses in delivering higher-risk customers.

Recommendations

The findings from the study indicate that the following recommendations should be implemented by companies offering traditional and subscription-based food delivery services:

Improve Customer Experience:

To improve customer experience, companies need to enhance their overall user experiences, including application performance, keeping up with delivery times, and making sure that they have responsive customer service.

Implement Loyalty Programs:

Implementing loyalty and/or reward programs can help companies engage with their customers and keep them as subscribers longer.

Use Predictive Analytics:

Employing predictive churn models to determine which customers are likely to cancel their subscriptions is a very important proactive measure for companies as they can implement retention strategies for those specific customers.

Personalize Customer Engagement:

Personalizing the customer experience through tailored promotions, such as promotional offers and product recommendations, may lead to greater customer satisfaction and a reduction in churn.

Monitor Customer Feedback:

Regularly analyzing customer feedback and service ratings will enable companies to identify and correct any issues prior to customers terminating their subscriptions.